

# Second Law of Thermodynamics

## 3. Entropy

Entropy is an extensive thermodynamic property that is measure of a system's thermal energy per unit temperature that is unavailable for doing useful work. Thermodynamic entropy is a non-conserved state function. For Isolated systems entropy never decreases. In statistical mechanics, entropy is a measure of the number of ways in which a system may be arranged, often taken to be a measure of 'disorder' (the higher the entropy the higher the disorder).

The infinitesimal change in the entropy ( $dS$ ) of a system is the infinitesimal transfer of heat energy ( $\delta Q$ ) to a closed system driving a reversible process, divides by temperature ( $T$ ) of the system.

$$\Delta S = \int \frac{\delta Q}{T}$$

It has unit Joule/Kelvin or  $dS = \int \frac{\delta Q}{T}$

Law of thermodynamics and entropy:

According to first law of thermodynamics

$$\delta Q = dU + PdV$$

From definition of Entropy

$$\Delta S = \frac{\delta Q}{T} \Rightarrow \delta Q = T \Delta S \Rightarrow TdS$$

$$TdS = dU + pdV$$

### Inequality of Clausius

Consider an irreversible cyclic engine working between  $T_1$  and  $T_2$ . If reversible engine is operating between same temperature then from Carnot theorem.

Efficiency of irreversible ( $\eta_{ir}$ ) will always smaller than efficiency of reversible engine ( $\eta_r$ ).

$$\eta_{irr.} < \eta_{rev.}$$

$$\frac{Q_1^{irr} - Q_2^{irr}}{Q_1^{irr}} < \frac{Q_1^{rev} - Q_2^{rev}}{Q_1^{rev}} \Rightarrow 1 - \frac{Q_2^{irr}}{Q_1^{irr}} < 1 - \frac{Q_2^{rev}}{Q_1^{rev}}$$

$$1 - \frac{Q_2^{irr}}{Q_1^{irr}} < 1 - \frac{T_2}{T_1} \Rightarrow \frac{Q_2^{irr}}{T_2} > \frac{Q_1^{irr}}{T_1}$$

F<sub>0</sub> irreversible cyclic Engine

$$\frac{Q_1^{irr}}{T_1} - \frac{Q_2^{irr}}{T_2} < 0$$

or  $\oint \frac{\delta Q}{T} < 0$

This relation is known as inequality of Clausius.

**Example:** Two Carnot Engines A and B are operated in series. The first one A receives heat at 900 K and rejects to a reservoir at temperature  $T$  K. The second engine, B, receives the heat rejected by the first Engine and in then rejects to a heat reservoir at 400 K.

Calculate the temperature  $T$  for the situation.

(a) The work outputs of the two engines are equal

(b) The efficiency of the two engines are equal.

For Engine A take in heat  $Q_1$  at temperature  $T_1$  and rejected heat  $Q$  at temperature  $T$ ; and the engine B taken in heat  $Q$  at temperature  $T$  and reject heat  $Q_2$  at temperature  $T_2$ .

**Solution:** (a)  $W_A = Q_1 - Q$      $W_B = Q - Q_2$

$$W_A = W_B \quad Q_1 - Q = Q - Q_2$$

$$\frac{Q_1 - Q_2}{Q} = 2$$

$$\frac{Q_1}{Q} = \frac{T_1}{T} \quad \text{and} \quad \frac{Q_2}{Q} = \frac{T_2}{T}$$

$$\frac{Q_1 + Q_2}{Q} = \frac{T_1 + T_2}{T} \Rightarrow \frac{T_1 + T_2}{T} = 2 \quad T = 650 \text{ K}$$

**Solution:** (b)  $\eta_A = \eta_B$

$$1 - \frac{T}{T_1} = 1 - \frac{T_2}{T}$$

$$T = (T_1 \cdot T_2)^{\frac{1}{2}} \quad T = 600 \text{ K}$$

**Example:** Calculate the charge in isothermal expansion from an initial volume  $V_i$  to volume  $V_f$

**Solution:** For reversible process

$$TdS = dU + PdV$$

For isothermal process  $dU = 0$

$$dS = \frac{PdV}{T}$$

$$P = \frac{nRT}{V}$$

$$dS = nR \int_{V_i}^{V_f} \frac{dV}{V} = nR \ln \frac{V_f}{V_i}$$

**Example:** A mass of liquid at a temperature  $T_1$  is mixed with an equal mass of the same liquid at a lower temperature  $T_2$ . the system is thermally insulated.

- (a) compute the entropy-change  
(b) show that it is necessarily positive.

**Solution:** Let  $c$  be the specific heat of the liquid. On the mixing equal mass  $m$  of the same liquid at temperature  $T_1$  and  $T_2$  let ( $T_1 > T_2$ ).

Let  $T$  be the equilibrium temperature at minute.

$$mc(T_1 - T) = mc(T - T_2)$$

$$T = \frac{T_1 + T_2}{2}$$

$$\Delta S_1 = \int_{T_1}^T \frac{\delta Q}{T} \text{ Entropy change of hotter liquid to cool from } T_1 \text{ to } T$$

$$\Delta S_1 = mc \int_{T_1}^T \frac{dT}{T} = mc \ln \frac{T}{T_1}$$

$\Delta S_2 \Rightarrow$  Entropy change of hotter liquid to heat from  $T_2$  to  $T$

$$\Delta S_2 = mc \ln \frac{T}{T_2}$$

$\Delta S$  = Entropy change of the system.

$$\Delta S = \Delta S_1 + \Delta S_2$$

$$= mc \ln \frac{T^2}{T_1 T_2} = mc \ln \left( \frac{\frac{T_1 + T_2}{2}}{\sqrt{T_1 T_2}} \right)^2 = 2mc \ln \frac{\frac{T_1 + T_2}{2}}{\sqrt{T_1 T_2}}$$

(b) we know arithmetic mean is greater than geometric mean.

$$AM > GM.$$

$$\text{So, } \frac{T_1 + T_2}{2} > \sqrt{T_1 T_2} \quad \text{So} \quad \Delta S > 0$$

**Example:** Compute the change in entropy when ice melt into steam. It is given that  $L_1$  is latent heat of fusion,  $c$  is specific heat at water and  $L_2$  latent heat at vaporization.

**Solution:** Assume  $T_1$  be the Kelvin temperature at which ice melts into water and  $T_2$  the Kelvin temperature at which water is boiled to steam.

$\Delta S_1$  is entropy change when ice is converted into water

$$\Delta S_1 = \frac{mL_1}{T_1}$$

$\Delta S_2$  Entropy change when water is heated from  $T_1$  to  $T_2$

$$\Delta S_2 = mc \int_{T_1}^{T_2} \frac{dT}{T}$$

$$\Delta S_2 = mc \ln \frac{T_2}{T_1}$$

$\Delta S_3$  Entropy change when water change into vapours

$$\Delta S_3 = \frac{mL_2}{T_2}$$

$$\text{Total change in entropy } \Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3 = \frac{mL_1}{T_1} + mc \ln \frac{T_2}{T_1} + \frac{mL_2}{T_2}$$